



POLITECNICO
MILANO 1863

Politecnico di Milano

Prova finale: Introduzione all'analisi di missioni spaziali
AA 2023-2024

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Data di consegna: 28/11/2023

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Table of symbols

CP:	change of plane	
CPA:	change of pericenter argument	
BT (pa):	bitangent transfer pericenter to apocenter	
BT (ap):	bitangent transfer apocenter to pericenter	
BE:	bielliptic transfer	
θ :	true anomaly	[rad]
Ω :	right ascension of ascendent node	[rad]
ω :	argument of the pericenter	[rad]
i :	inclination of the orbital plane	[rad]
a :	semi-major axis	[km]
e :	eccentricity	
p :	parameter	[km]
h :	specific angular momentum	[km ² /s]
MEO:	Medium Earth Orbit	
Δv :	delta of velocity	[km/s]
Δt :	delta of time	[s]

As the considered orbits are Earth-focused, the gravitational parameter used is: $\mu = 398600 \text{ km}^3/\text{s}^2$

1. Introduction

The aim of the following report is to evaluate different orbital strategies, to move a satellite from an initial to a final point on two different given orbits.

The team developed three strategies. The first strategy is a combination of standard maneuvers, whereas the other two are non-conventional alternatives.

In the first stage of the research all the members of the team worked together to write all the functions and scripts necessary to perform the calculation on MATLAB.

In the second stage the team examined different possibilities of executing the orbital transfer and all the pros and cons for every maneuver.

The report is divided into four main sections:

- the first section revolves around the characterisation of the initial orbit: given the cartesian coordinates of the starting point, the team shifted to the Keplerian parameters of the orbit;
- the second section focuses on the final orbit: the team calculated the position and velocity of the satellite in the geocentric equatorial system from the given Keplerian parameters of the final point;
- in the third section different transfer strategies are evaluated;
- in the fourth section the team compared all the prospects to choose the strategy offering the best trade-off in terms of efficiency and time.

2. Initial orbit characterisation

2.1 Initial orbit parameters

The parameters of the initial orbit were given in the geocentric equatorial reference system:

	Initial point position [km]	Initial point velocity [km/s]
X	-7244.1296	2.1140
Y	-4751.0447	-4.2990
Z	-465.5401	-4.7890

Using the software MATLAB, then the team calculated the Keplerian parameters of the initial orbit:

a [km]	8664.3	Ω [rad]	0.5268
e	0.1249	ω [rad]	1.5111
i [rad]	0.7864	θ [rad]	1.7064

2.2 Other parameters

Having all the necessary data it was possible to extract some secondary parameters of the initial orbit:

Perigee radius [km]	7582.1	Perigee velocity [km/s]	7.6901
Apogee radius [km]	9746.6	Apogee velocity [km/s]	5.9823
Energy [km^2/s^2]	-23.0	p [km]	8529.2
Period [s]	8026.3	h [m^2/s]	58307.1

2.3 Description and graphical representation

The orbit can be considered a MEO, as seen from the calculated semi-major axis and perigee and apogee radius. The energy of the orbit is negative, as expected from theoretical results for elliptical orbits. The high value of the inclination ($\sim 45^\circ$) means the orbit is highly inclined to the equator, as shown in *Figure 1* below:

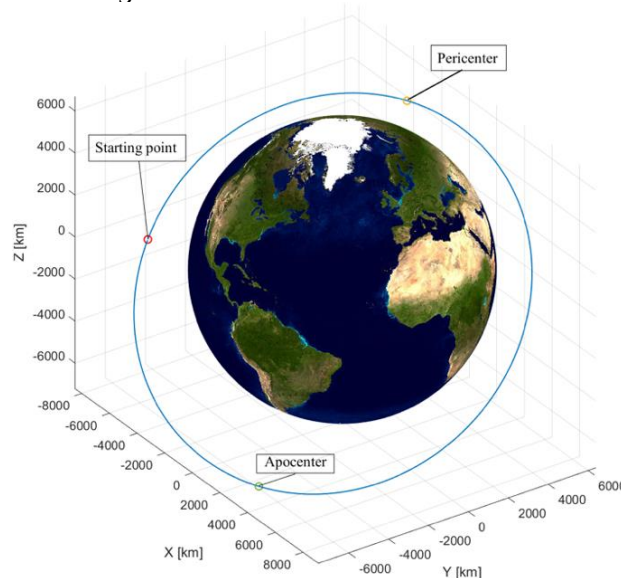


Figure 1: Initial Orbit

3. Final orbit characterisation

3.1 Final orbits parameters

The parameters of the final orbit were given in the perifocal reference system:

a [km]	14250.0	Ω [rad]	0.3314
e	0.3205	ω [rad]	1.4320
i [rad]	0.9687	θ [rad]	2.9690

Using the software MATLAB, then the team calculated the Cartesian coordinates of the final orbit:

	Initial point position [km]	Initial point velocity [km/s]
X	-2135.7	3.6195
Y	-11388.9	0.3691
Z	-14659.6	-1.2058

3.2 Other parameters

Having all the necessary data it was possible to extract some secondary parameters of the final orbit:

Perigee radius [km]	9682.9	Perigee velocity [km/s]	7.3728
Apogee radius [km]	18817.1	Apogee velocity [km/s]	3.7939
Energy [km ² /s ²]	-13.986	p [km]	12786.2
Period [s]	16929.1	h [m ² /s]	71390.4

3.3 Description and graphical representation

The final orbit is a MEO too. The inclination has increased ($\sim +10^\circ$), as well as the eccentricity. Having a different inclination from the initial orbit, a maneuver is required to change the orbital plane, in order to reach the final point. The final orbit can be visualized in Figure 2:

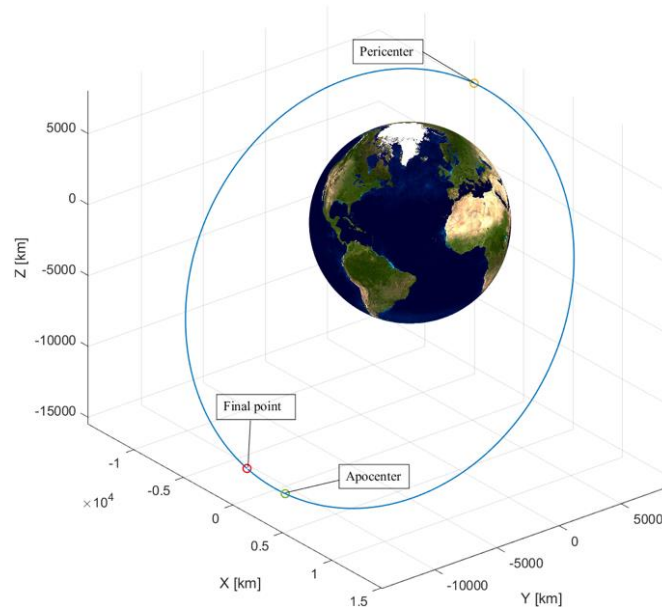


Figure 2: Final Orbit

4. Transfer trajectory definition and analysis

4.1 General description

Initial and final orbits are different in shape, orbital plane inclination and orientation. Therefore, it will be necessary to execute multiple maneuvers to achieve the final position and velocity. Three different strategies have been studied from the point of view of time taken and propellant used, and are presented hereafter:

1. Standard strategy
Change of plane – change of orientation – change of shape (bitangent transfer)
2. Alternative Strategy 1
Change of plane – change of orientation – change of shape (at the point of tangency)
3. Alternative Strategy 2
Change of plane – change of orientation – change of shape (secant transfer)

4.2 Transfer strategies

4.2.1 Standard strategy

The team analyzed all the different combinations of the following standard maneuvers: change of plane, change of perigee anomaly, bitangent transfer, bielliptic transfer. The results are shown in *Figure 3*. The most convenient one turned out to be the one composed, in this order, by a bitangent transfer, a change of plane and a change of anomaly of the pericenter.

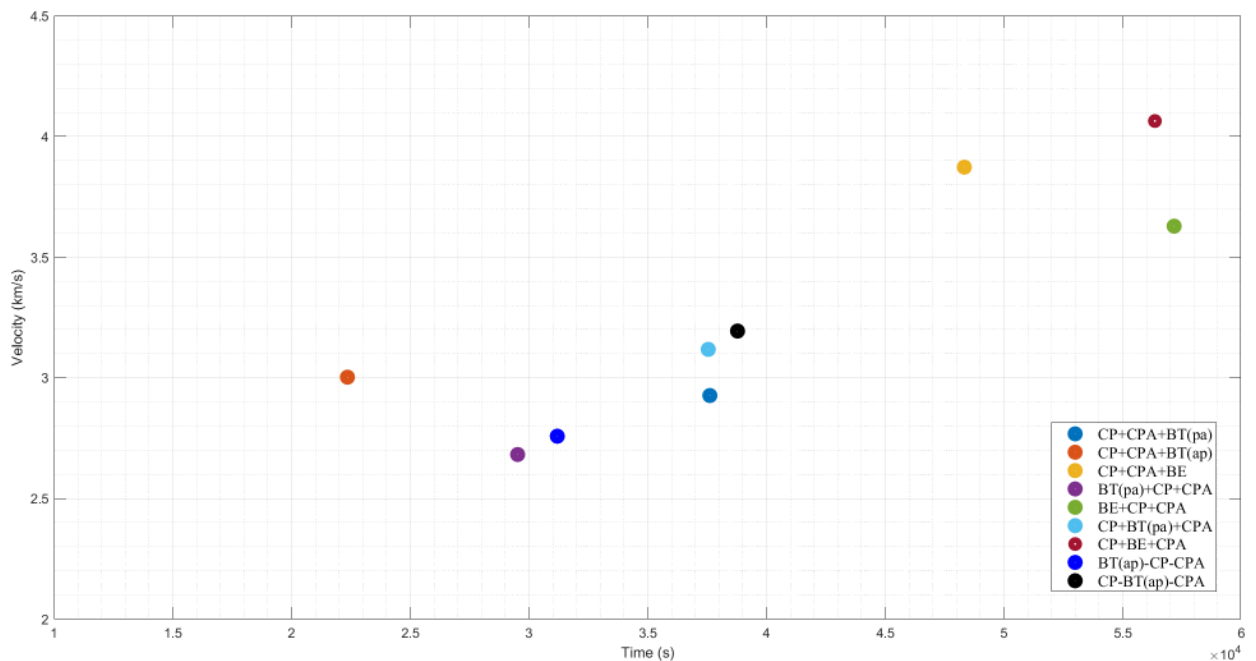


Figure 3: Standard Transfer Options

The chosen strategy (*Figure 4*) was selected as it gives the best trade-off between Δv and Δt , having given priority to saving propellant.

As mentioned, the first maneuver of this strategy is a bitangent transfer (P-A).

Firstly, the satellite must move from the release point to the perigee of the initial orbit:

$$\Delta t = 6166 \text{ s.}$$

There, it moves on a bitangent transfer orbit to reach an orbit the same shape as the final one:

$$[a_i, e_i, i_i, \Omega_i, \omega_i, 0] \rightarrow [a_f, e_f, i_i, \Omega_i, \omega_i, \pi] \quad \Delta v = 1.2726 \text{ km/s} \quad \Delta t = 7546 \text{ s}$$

The second maneuver is a change of plane. Once the correct anomaly for it has been reached, and the change of plane executed, the satellite will be on an orbit with same shape, inclination and RAAN as the final one:

$$[a_f, e_f, i_i, \Omega_i, \omega_i, \theta_1] \rightarrow [a_f, e_f, i_f, \Omega_f, \omega_2, \theta_1] \quad \Delta v = 1.0438 \text{ km/s} \quad \Delta t = 3906 \text{ s}$$

Last step of the strategy is to align the eccentricity axis of this intermediate orbit to the one of the final orbit. To do so, a third and last maneuver is necessary. The change of anomaly of the pericenter can be done once the satellite has reached a true anomaly $\theta_2 = \Delta\omega/2$:

$$[a_f, e_f, i_f, \Omega_f, \omega_2, \theta_2] \rightarrow [a_f, e_f, i_f, \Omega_f, \omega_f, \theta_3] \quad \Delta v = 0.3657 \text{ km/s} \quad \Delta t = 4423 \text{ s}$$

Once the satellite is on the final orbit, it just has to move to reach the final point: $\Delta t = 7478 \text{ s.}$

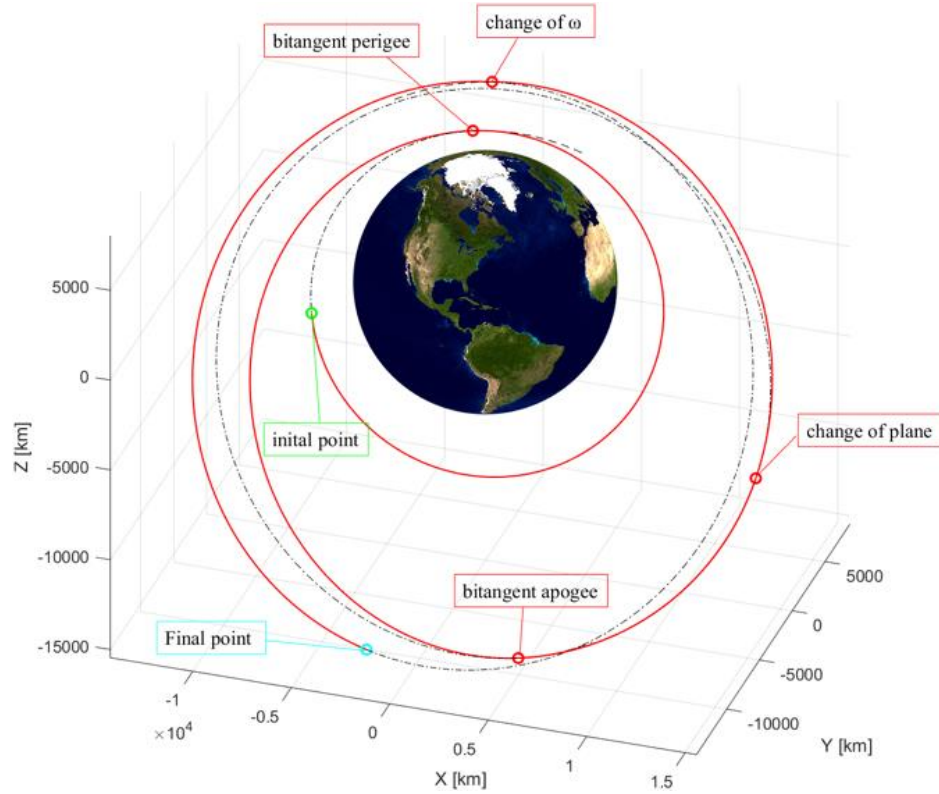


Figure 4: Standard Transfer

4.2.2 Alternative Strategy 1

The first non-conventional strategy (Figure 5) developed by the team consists in three maneuvers performed in the following order:

- change of plane;
- change of pericenter anomaly, until the transfer orbit is tangent with the final one;
- change of shape at the point of tangency.

The first maneuver once the satellite reaches the correct true anomaly, is a change of plane to bring the initial orbit on the same plane of the final one:

$$[a_i, e_i, i_i, \Omega_i, \omega_i, \theta_i] \rightarrow [a_i, e_i, i_f, \Omega_f, \omega_1, \theta_1] \quad \Delta v = 1.4797 \text{ km/s} \quad \Delta t = 3532 \text{ s}$$

The second step is to identify a transfer orbit with the same a, e, i, Ω of the orbit after the change of plane that shares a tangency point with the final one. According to this, the following system has been resolved in *MATLAB*:

$$\begin{cases} \frac{p_2}{1 + e_2 \cos \theta_2} = \frac{p_3}{1 + e_3 \cos \theta_3} \\ \frac{e_2 \sin \theta_2}{1 + e_2 \cos \theta_2} = \frac{e_3 \sin \theta_3}{1 + e_3 \cos \theta_3} \end{cases}$$

Where θ_2, θ_3 are the points of tangency on the transfer orbit and on the final one, respectively.

To reach the transfer orbit a change of anomaly of the pericenter is necessary:

$$[a_i, e_i, i_f, \Omega_f, \omega_1, \theta_1] \rightarrow [a_i, e_i, i_f, \Omega_f, \omega_2, \theta_2] \quad \Delta v = 1.7005 \text{ km/s} \quad \Delta t = 1056 \text{ s}$$

The third maneuver is a change of shape at the point of tangency:

$$[a_i, e_i, i_f, \Omega_f, \omega_2, \theta_2] \rightarrow [a_f, e_f, i_f, \Omega_f, \omega_f, \theta_3] \quad \Delta v = 1.3480 \text{ km/s} \quad \Delta t = 2042 \text{ s}$$

The transfer ends when the satellite arrives in θ_f : $\Delta t = 7429 \text{ s}$.

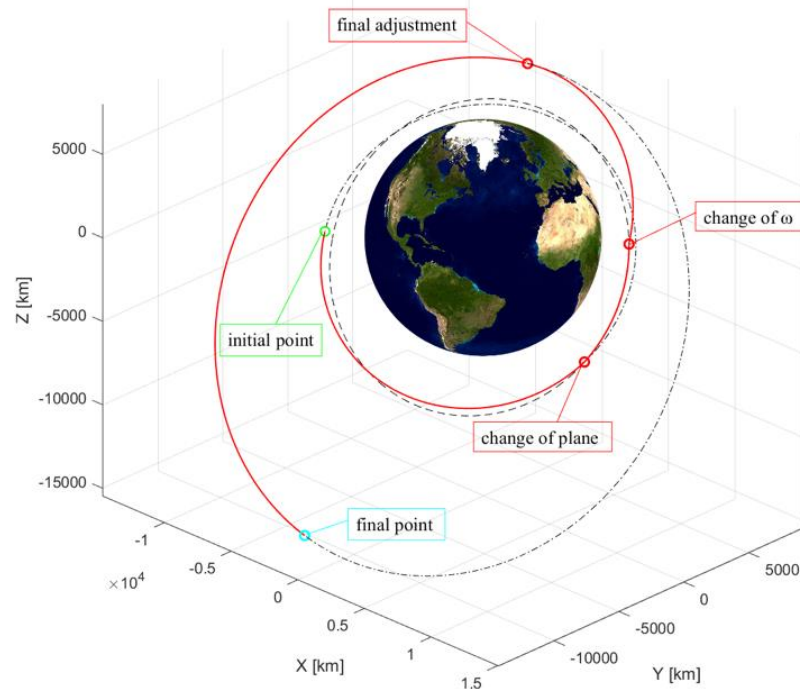


Figure 5: Alternative Strategy 1

4.2.3 Alternative Strategy 2

The second strategy (Figure 6) still composes of three maneuvers:

- change of plane;
- change of the anomaly of pericenter;
- change of shape using a secant orbit.

The third maneuver is a non-conventional change of shape. The team chose a strategy like this to minimize the time required, at the expense of the propellant used. It is in fact a secant maneuver, which allows the satellite to shorten the path between orbits.

The first maneuver is just like the one described in the first alternative strategy:

$$[ai, ei, ii, \Omega i, \omega i, \theta i] \rightarrow [ai, ei, if, \Omega f, \omega 1, \theta 1] \quad \Delta v = 1.4797 \text{ km/s} \quad \Delta t = 3532 \text{ s}$$

The next step is to execute a change of anomaly of pericenter, to align the eccentricity axis of the orbit the satellite is now on, with the final orbit. To do so, it is necessary to wait until the true anomaly reaches a value of $\Delta\omega/2$.

$$[ai, ei, ii, \Omega i, \omega 1, \theta 1] \rightarrow [ai, ei, if, \Omega f, \omega f, \theta 2] \quad \Delta v = 0.1745 \text{ km/s} \quad \Delta t = 2533 \text{ s}$$

After having changed plane and anomaly of pericenter, the satellite moves on an auxiliary orbit. This auxiliary orbit is designed to have its pericenter exactly on the anomaly where the change of anomaly of pericenter was executed. The same pericenter of the auxiliary orbit is also the starting point of this last maneuver. The final point is the intersection point of the auxiliary and the final orbit. This way the satellite has only travelled a relatively short time on the auxiliary orbit, despite consuming lot of propellant just for this maneuver.

$$[ai, ei, if, \Omega f, \omega f, \theta 2] \rightarrow [af, ef, if, \Omega f, \omega f, \theta f] \quad \Delta v = 1.5011 \text{ km/s} \quad \Delta t = 6209 \text{ s}.$$

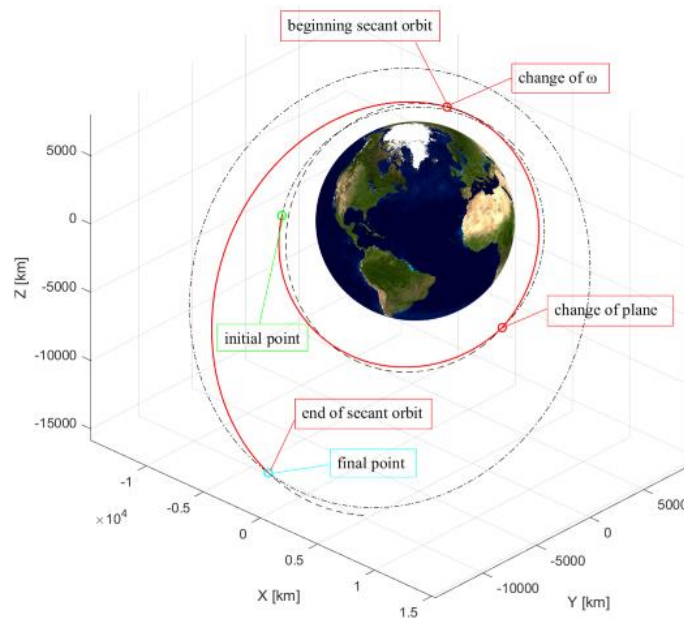


Figure 6: Alternative Strategy 2

5. Conclusions

The three strategies analysed in the report are now compared in *Figure 7*:

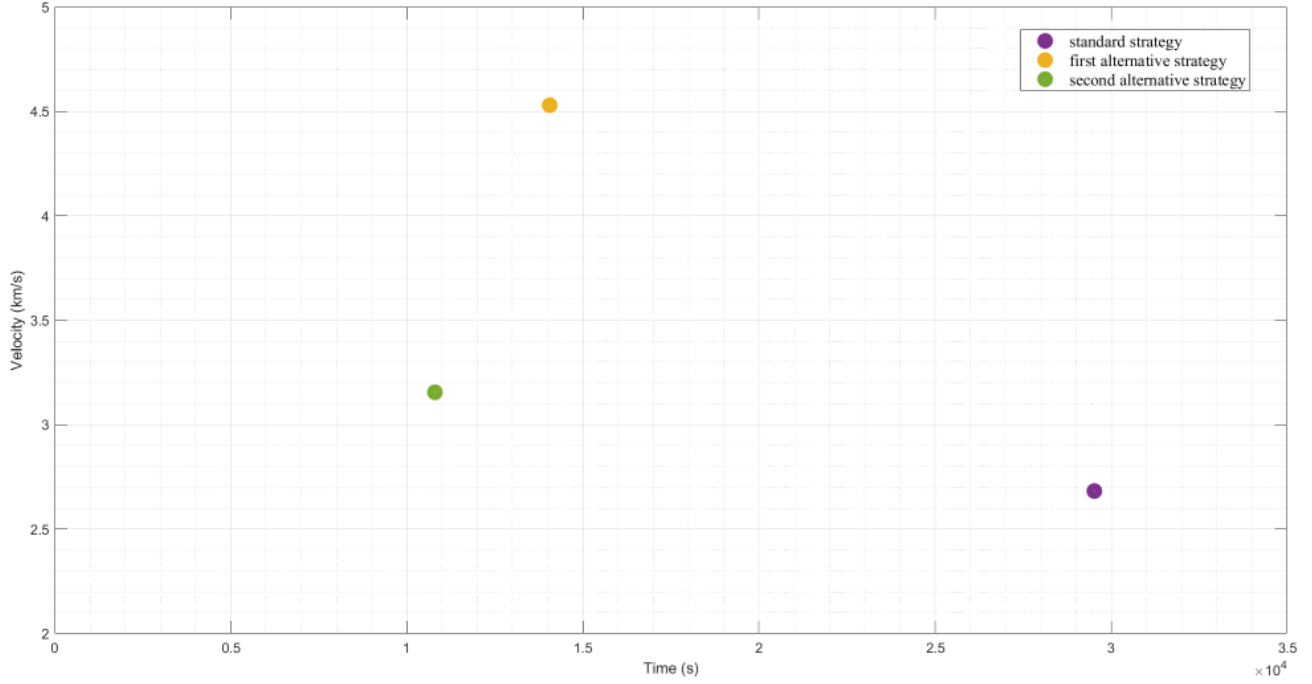


Figure 7: Strategies Comparison

In terms of Δv , the standard transfer is the most convenient (2.6822 km/s); it allows to save more fuel, since the two quantities are related by the Tsiolkovsky equation¹. In addition, less fuel means that this strategy is the cheapest one but it requires a very high transfer time (29521 s).

The second strategy decreases by 50% the Δt required but ΔV increases a lot compared to the standard one (4.5281 km/s), so the maneuver cost is very high.

The third strategy is the best trade-off between the two quantities. In fact transfer time is minimized (12274 s) while ΔV slightly increased compared to the first strategy (3.1552 km/s).

¹Tsiolkovsky equation: $\Delta v = I_{sp} g_0 \ln \frac{m_i}{m_f}$

6. Appendix

Transfer 1 (standard strategy)

t (s)	a (km)	e (-)	i (rad)	Ω (rad)	ω (rad)	θ (rad)	Δv (km/s)
0	8664.3	0.1249	0.7864	0.5269	1.5110	1.7064	-
6166	8664.3	0.1249	0.7864	0.5269	1.5110	0.0000	0.9670
	13200	0.4256	0.7864	0.5269	1.5110	0.0000	
13712	13200	0.4256	0.7864	0.5269	1.5110	3.1415	0.3056
	14250	0.3205	0.7864	0.5269	1.5110	3.1415	
17619	14250	0.3205	0.7864	0.5269	1.5110	4.0173	1.0439
	14250	0.3205	0.9687	0.3314	1.6367	4.0173	
22043	14250	0.3205	0.9687	0.3314	1.6367	6.1808	0.3657
	14250	0.3205	0.9687	0.3314	1.4320	0.1023	
29521	14250	0.3205	0.9687	0.3314	1.4320	2.9690	-

Transfer 2

t (s)	a (km)	e (-)	i (rad)	Ω (rad)	ω (rad)	θ (rad)	Δv (km/s)
0	8664.3	0.1249	0.7864	0.5269	1.5110	1.7064	-
3532	8664.3	0.1249	0.7864	0.5269	1.5110	4.0173	1.4797
	8664.3	0.1249	0.9687	0.3314	1.6367	4.0173	
4587	8664.3	0.1249	0.9687	0.3314	1.6367	4.8008	1.7005
	8664.3	0.1249	0.9687	0.3314	-1.3280	1.4823	
6629	8664.3	0.1249	0.9687	0.3314	-1.3280	2.9001	1.3480
	14250	0.3205	0.9687	0.3314	1.4320	0.1401	
14056	14250	0.3205	0.9687	0.3314	1.4320	2.9690	-

Transfer 3

t (s)	a (km)	e (-)	i (rad)	Ω (rad)	ω (rad)	θ (rad)	Δv (km/s)
0	8664.3	0.1249	0.7864	0.5269	1.5110	1.7064	-
3532	8664.3	0.1249	0.7864	0.5269	1.5110	4.0173	1.4797
	8664.3	0.1249	0.9687	0.3314	1.6367	4.0173	
6065	8664.3	0.1249	0.9687	0.3314	1.6367	6.1808	0.1745
	8664.3	0.1249	0.9687	0.3314	1.4320	0.1024	
12274	13406	0.4341	0.9687	0.3314	1.5344	0.0000	1.5011
	13406	0.4341	0.9687	0.3314	1.4320	2.8666	
12274	14250	0.3205	0.9687	0.3314	1.4320	2.9690	-